

Script lesson 3: Writing a literature review, learning how to paraphrase and how to use tenses in articles

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1. Objectives

- Outline the research story of your thesis / article
- Discuss tense use in research articles and theses
- Examine the literature review section of articles / theses
- Focus on some ways of reporting research and using reporting verbs
- Examine tense use in citations and paraphrasing
- Learn how to paraphrase
- Practise paraphrasing
- Do exercises on paraphrasing and formal expression, comparing your answers with examples or possible answers

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2. Outlining the research story of your thesis / article

Writing a research article or thesis is essentially like writing a story. This plot line describes researchers' journey through the scientific method, describing their attempts to tackle a question step by step. The main characters of these stories encounter a problem (Introduction, Literature review), develop ideas about the problem (Hypothesis), or expose gaps that are still necessary to solve the problem (gap statement), or ask some questions to guide solving the problem (Research Questions). Thereafter, they develop ways (Methods) to find out the answers to the problems (Results/Findings), before drawing conclusions (Discussion, Conclusion). All the while they show who helped them along the way (References).

FAIRY TALES	SCIENTIFIC PAPERS
<i>Once upon a time (background)</i>	Introduction: Why did we start?
<i>The test</i>	Methods: What did we do?
<i>The results</i>	Results: What did we find?
<i>The moral</i>	Discussion: What does it mean? (the message)

The message drives the story, but you cannot write the message until you have finished your analysis and decided on the implications of your main finding. Since you cannot start writing the story before you know those main conclusions and implications, where do you start? Researchers often start their theses or research articles by writing the Methods if they have their research protocols ready. However, when that is not the case, the Literature Review part of the Introduction section is a good place to start, as you need to read a lot in preparation, so you can feed those notes into your thesis writing. Of course, you cannot include all of that reading into your Literature review: you would sift through everything and include only the most pertinent sources and conclusions.

The slides illustrate the use of the outlining function in WORD to build the outline of your paper that you fill in with paragraphs as you read and write.

Different verb tenses are used in various sections of a research manuscript; therefore, the next section will focus on these tenses.

3. Tense use in research articles

The publication *Manual of the American Psychological Association* makes the following recommendations regarding tense use in research articles:

USE PAST TENSE . . .

- **To describe your methodology and report your results.** At the time you write your report, thesis, dissertation, or article, you have completed your study, so should use *past tense* in your methodology section to record what you did, and in your results section to report what you found. Consider these examples:
“We hypothesized that adults would remember more items than children.”
“We extracted tannins from the leaves by bringing them to a boil in 50% methanol.”

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- **When referring to the work of previous researchers.** When citing previous research, use past tense. Whatever a previous researcher said, did or wrote happened in the past. Results relevant only in the past, or to a particular study and not yet generally accepted should also be expressed in past tense. Consider these examples:

“Smith (2008) reported that adult respondents remembered 30 percent more than children.

“(Smith's study was completed in the past and his finding was specific to that particular study.)

However, when the findings have become generally accepted as true, use the present tense to refer to the finding: “Previous research showed that children confuse the source of their memories more often than adults.” (Research was conducted in the past, but the finding is now widely accepted.)

- **To describe a fact, law or finding that is no longer considered valid and relevant.** Look at these examples:
“Nineteenth-century physicians held that women got migraines because they were ‘the weaker sex’, but current research shows that the causes of migraine are unrelated to gender.” (Note the shift here from past tense [discredited belief] to present [current belief].)

USE PRESENT TENSE . . .

- **To express findings that continue to be true.** Use present tense to express truths or facts supported by research results that won't change, as in the following examples:
“Genetic information is encoded in the sequence of nucleotides on DNA.”
“Galileo asserted that the earth revolves the sun.” (The asserting took place in the past, but the earth is still revolving around the sun.)
“We chose Vietnam for this study because it has a long coastline.” (Use past tense to indicate what you did [chose Vietnam], but present tense to indicate the length of Vietnam's coastline is unlikely to change.)
- **To refer to the article, thesis or dissertation itself.** Use the present tense in reference to the thesis or dissertation itself and what it contains, shows, etc. For example:
“In Table 3, I show that the main cause of weight increase was nutritional value of the feed.” (Table 3 will always show this; and will be true whenever anyone reads this sentence.)
- **To discuss your findings and present your conclusions. Also use present tense to discuss your results and their implications.** Consider these examples:
“Weight increased as the nutritional value of feed increased. These results suggest that feeds higher in nutritional value contribute to greater weight gain in livestock.” (Use past tense to indicate what you found [weight increased], but present tense to suggest what result implies.)

You could also refer to the infographic below for an overview.

Scientific Writing: a Verb Tense Review

1 Abstract Past To talk about actual results. Present To talk about general facts, the paper itself or analysis of findings.	2 Introduction Present To talk about general background information. Present Perfect To talk about previous research.	3 Methods Past To talk about what you did. TIP: passive voice is common. Present To explain diagrams/figures. Ex: Table 1 shows...
PAST To talk about events that have been completed. Ex1: We found that... Ex2: Protocol X was followed. [Passive voice] PRESENT PERFECT To talk about events that started in past but are still ongoing or recently completed. Ex: Many experiments have focused on...		PRESENT To talk about events that are general fact, discuss current meanings, and suggest future applications. Ex1: Insulin and glucagon regulate blood glucose levels. Ex2: Increased hormone production indicates...
4 Results Past To talk about actual results. Present To explain diagrams/figures. Ex: Table 1 shows...	5 Discussion Present To interpret and talk about significance of findings. Past To briefly summarize findings.	6 Conclusion Past To refer to your completed research. Present To talk about implications and suggest future research.



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Note that the present perfect tense in the Introduction is used mainly for research that was started in the past but is most probably still continuing.

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4. The Literature Review and study background

Purposes

Consider some articles/theses from your field of study.

- Where does one find references to/information about the study background and other sources used?
- Write down at least four purposes of a study background and literature review?

Consider the following purposes of including a literature review in a thesis (Murray 2002:101-108):

- Learning about available research and literature on the topic (provide a context), and incorporating it into the thesis argument,
- Situating the research in a broader context,
- Demonstrating what the writer knows already,
- Summarizing and evaluating other people's work,
- Exploring what the schools of thought in the literature are, and what the 'great debates' in the area are,
- Identifying gaps in existing research,
- Developing an understanding of theory and practice.

Along these lines, Murray (2002:108) also points out some questions that a literature review should answer.

It is important to remember that you as the writer of the thesis play an active role in the literature review, as you are **interpreting, synthesizing and criticizing** research that is relevant to your research topic.

During your literature search, you keep track of sources, their main ideas, and your opinions of their ideas.

While the background of a study and the literature review both form part of the introduction of an article, thesis and proposal, there are important differences between them, as summarized by the following infographic by *Editage Insights*.

Thereafter, we focus on paraphrasing skills, as they are particularly important in these sections of a research manuscript.

6 Differences between a study background and a literature review		
	Background of a study 	Literature review 
Q Where does it appear in the research paper?	In the first part of the Introduction section	After the background in the Introduction section
Q What does it present?	The context of the study	A critical analysis of existing literature
Q What is its purpose?	To highlight the significance of the study	To evaluate the progress of knowledge in the research area
Q What does it do?	Introduces the topic in context and explains how it leads up to the research problem	Helps identify the gaps in literature that led to the study
Q What does it involve?	Introducing the broad area of research and narrowing it down to the problem at hand	Analyzing the literature to show what research has been done on the topic
Q What is its length?	Is short and concise	Is lengthy and detailed
 www.editage.com Your partner in research publication     www.editage.com/insights Resources throughout your publication journey		

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3.1 Paraphrasing

In the background and literature review sections, you will need to **summarize and paraphrase** writers' words in your own words. Remember that only changing a few words of the original sentence does not constitute a good paraphrase. One needs to change both the words and the sentence structure of the original while staying true to the main ideas of the text.

Paraphrasing is a useful skill to learn because it helps one really to understand a text in depth before putting it in one's own words. To paraphrase correctly, one can follow these steps:

- **Read** the original text thoroughly twice and **identify the main ideas**.
- Write the ideas in your **own words**. You do not need to use synonyms for **technical and specialist terminology** or **short expressions**. For example, terms such as *Neglected Tropical Diseases* or *Mycobacterium tuberculosis (Mtb)* are specialist terms for which you should not try to use synonyms, while abbreviations such as *DNA* are very well-known terms and should be used as such. Include enough support and detail so that your message is clear.
- Make sure the paraphrase summary reads smoothly (that the text is **cohesive and coherent**). Focus on old-to-new information flow; use transition devices where necessary; and provide supporting detail. You do not want a collection of sentences that do not flow.
- If it is impossible to use your own words, then quote the material. Remember, however, that too much quoting will not likely result in a successful paper. Information from sources should *support, but not replace*, your own ideas, interpretations, and explanations.
- **Reference** the source material.

The next section focuses on plagiarism and tense use when referring to sources. Further on, Section 5.1 of this script contains paraphrasing exercises, while Section 5.2 includes exercises on formal and informal writing in 5.2.

For further reading on **paraphrases, quotations and summaries**, please see the Purdue Online Writing Lab's pages here: <https://owl.english.purdue.edu/owl/resource/563/01/> and <https://owl.english.purdue.edu/owl/resource/563/02/>

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3.2 Plagiarism

In the literature review section of an article, you should especially be aware of **plagiarism**: whenever you refer to someone else's work or research, you should be sure to reference the source. According to plagiarism.org (<http://www.plagiarism.org/plagiarism-101/what-is-plagiarism>), all of the uses below are forms of plagiarism:

- turning in someone else's work as your own
- copying words or ideas from someone else without giving credit
- failing to put a citation (quotation) in quotation marks
- giving incorrect information about the source of a quotation
- changing words but copying the sentence structure of a source without giving credit
- copying so many words or ideas from a source that it makes up the majority of your work, whether you give credit or not

3.3 Tense use when referring to sources

When one considers the way previous research and publications are reviewed, a variety of patterns of use emerge. The simple past tense, present perfect tense and present tense are all used, depending on the situation. The following patterns emerge:

Past tense, with the researcher activity as the agent

- Huang (2007) *investigated* the causes of airport delays.
- The causes of airport delays *were investigated* by Huang (2007). (Passive voice)

In pattern 1, the past tense is usually used to refer to single studies.

Present perfect tense

- The causes of airport delays *have been widely investigated* (Hyon, 2004; Huang, 2007).
- Several researchers *have studied* the causes of airport delays (Hyon, 2004; Huang, 2007).
- There *have been* several investigations into the causes of airport delays (Hyon, 2004; Huang, 2007).

In Pattern 2, the present perfect is used to refer to areas of inquiry, or to researcher activity over a period of time in the recent past, or to recent trends in research.

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Present tense

- The causes of airport delays are complex (Hyon, 2004; Huang, 2007).
- Airport delays appear to have a complex set of causes (Hyon, 2004; Huang, 2007).

In Pattern 3, the present tense refers to the current state of knowledge.

Now examine the article or thesis you have accessed or brought with you.

1. Highlight references to sources.
2. Write down at least 4 different verbs that are used to refer to sources (for example 'Rogers (2004) *concluded* that...').
3. Compare your list with those of the other students and make a list of useful verb phrases.

Note: In scientific articles, reporting verbs referring to sources are often omitted for the sake of conciseness, with the finding of the study summarized and referenced, for example:

- The causes of airport delays *have been widely investigated* (1, 2).

The following **reporting verbs** are commonly used in the literature review (especially in theses) when referring to the work of other authors and paraphrasing their words. In articles, which are much shorter, they are often omitted.

1. *Believe, consider, say, suggest, think* (often used with 'as'), for example:
'As Williams suggests, the first visit to the general practitioner is crucial.'
2. *Argue, assert, claim, conclude, confirm, contend, insist, maintain, state, warn*, e.g.:
'Williams maintains that xenophobic attitudes have had an influence on the increase in attacks on foreigners.'
3. *Demonstrate, explain, indicate, note, prove, reveal, show*, for example:
'As Williams demonstrated, cognitive behavioural therapy influences participants' decisions.'
4. *Accept, agree, deny, doubt, question*, for example:
'In her research on study habits, Smith doubts Williams's findings on the positive influence of cognitive behavioural therapy.'
5. *Blame, condemn, criticize* (to take a critical or negative position), for example:
'Shelley criticizes therapists for their lack of understanding of victims' traumatic experiences.'
6. *Describe, characterize, define, evaluate, identify, portray, present*, for example:
'In his studies in the field of anthropology, Nicholson often portrays these tribes as peace-loving.'

(Adapted from Paterson, 2013:143)

Try to use a variety of verbs to make your literature review more interesting.

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When you cite sources and write references in the bibliography make sure that you follow the system prescribed/preferred by your department (eg. Harvard style, Chicago style, the APA (American Psychological Association) style, Vancouver style), and be consistent – do not switch between different ways of referencing! For an overview of these styles, you could consult <http://www.newcastle.edu.au/service/library/tutorials/newcat/writing/page20.html> and http://www.chicagomanualofstyle.org/tools_citationguide.html.

Section 5 continues below.

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5.1 Paraphrasing examples and exercises

Read the following original text from the Purdue Owl website and 2 paraphrases of it. For comparison, there is a legitimate summary, a plagiarized version and a summarized version. (<https://owl.english.purdue.edu/owl/resource/619/1/> . A machine-paraphrased version is also included.

The original passage:

Students frequently overuse direct quotation in taking notes, and as a result they overuse quotations in the final [research] paper. Probably only about 10% of your final manuscript should appear as directly quoted matter. Therefore, you should strive to limit the amount of exact transcribing of source materials while taking notes. (Lester 1976:46-47).

A legitimate paraphrase:

In research papers students often quote excessively, failing to keep quoted material down to a desirable level. Since the problem usually originates during note taking, it is essential to minimize the material recorded verbatim (ibid).

A plagiarized version:

Students often use too many direct quotations when they take notes, resulting in too many of them in the final research paper. In fact, probably only about 10% of the final copy should consist of directly quoted material. So, it is important to limit the amount of source material copied while taking notes.

A machine-paraphrased version:

Understudies much of the time abuse direct citation in taking notes, and accordingly they abuse citations in the last [research] paper. Presumably just around 10% of your last composition ought to show up as straightforwardly cited matter. Subsequently, you ought to endeavour to confine the measure of precise deciphering of source materials while taking notes. (<http://paraphrasing-tool.com/>)

An acceptable summary:

Students should take just a few notes in direct quotation from sources to help minimize the amount of quoted material in a research paper (Lester 46-47).

Compare the different versions to each other. Identify the reasons why the plagiarized version is not acceptable. Also read the machine-paraphrased version. What you will notice is that the computer program maintains the syntax of the source material and simply replaces most words by a synonym, as if you used Thesaurus in Word and picked the first match for each word in your text. As a result, some words seem odd and out of context, and

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the overall meaning of the original is lost. This is because the computer program, unlike your brain, does not make sense of the text. Also, your lecturer will immediately be able to see that your paraphrase is not your own work!

Read the following texts and write a paraphrase of each, using the techniques above. The Purdue website (<https://owl.english.purdue.edu/owl/resource/619/2/>) provides more examples of texts with paraphrases, including some of the following ones.

1. "The Antarctic is the vast source of cold on our planet, just as the sun is the source of our heat, and it exerts tremendous control on our climate," [Jacques] Cousteau told the camera. "The cold ocean water around Antarctica flows north to mix with warmer water from the tropics, and its upwellings help to cool both the surface water and our atmosphere. Yet the fragility of this regulating system is now threatened by human activity." From "Captain Cousteau," Audubon (May 1990):17.

Paraphrase

2. Of the more than 1000 bicycling deaths each year, three-fourths are caused by head injuries. Half of those killed are school-age children. One study concluded that wearing a bike helmet can reduce the risk of head injury by 85 percent. In an accident, a bike helmet absorbs the shock and cushions the head. From "Bike Helmets: Unused Lifesavers," Consumer Reports (May 1990): 348.

Paraphrase

[Type here]

3. Arguably the most incredible modern architectural feat, the Burj Khalifa is the world's tallest building as it soars over 2,500 feet in the sky. Designed by Adrian Smith and engineered by Bill Baker, the multi-billion-dollar structure officially opened in 2010 and features the world's longest-travelling elevator and five million square feet of floor space. (<http://all-that-is-interesting.com/modern-architectural-feats>)

Paraphrase

4. 'It is widely agreed that men and women think and act in different ways. Women appear to have better memories, better social skills and are better at multi-tasking. Men, in contrast, seem to focus better on single issues and have superior motor and spatial skills, although clearly many people are exceptions to these patterns' (Bailey, 2015:63).

Paraphrase

5.2 Formal and informal language use exercises

Examine these sentences and paragraphs and **highlight** the instances of informal language use. Thereafter, rewrite the sentences in a more formal way, keeping in mind the characteristics of academic language outline above.

1. Up to now people haven't really looked much into the relationship between mobile phone use and the amount of sport people do.

[Type here]

2. You can use this model to make recycling of plastic more efficient.
3. So why did the financial crisis occur? There are many reasons.
4. There are several studies in epidemiology that have shown that when people don't drink too much, they have a lower risk of developing heart disease in comparison to those people who drink a lot of alcohol (Adapted from Swales and Feak 2012:26).

5. Assignment 3

The use of Artificial Intelligence tools are NOT permitted for this assignment.

Assignment 3 has two parts.

Part 1: Reviewed assignment and

Part 2: Reading and recording sources

1. Paraphrase and summarize 3 of the 6 paragraphs from epidemiological journal articles below. The summary paraphrase should be about one third to half the length of the original paragraph, but contain the essential information in the original text, stated in your own words.

Paragraph 1:

Extremely low-frequency magnetic fields are designated as possibly carcinogenic in humans based on an epidemiologic association with childhood leukaemia, though no biologic mechanism has been identified. In a national case-control study of childhood cancers in England and Wales, both a significant trend ($P < 0.01$) in risk of leukaemia with distance from power lines and a significant dose-response relationship comparing risk of leukaemia for those living <200 m and 200–600 m from a power line with those living >600 m were observed. These associations were, however, not fully explained by exposure to magnetic fields because no statistically significant excess risk was observed with increasing magnetic fields.

Paragraph 2:

There is growing epidemiological evidence that transportation noise affects health in various ways. Transportation noise induces sympathetic and endocrine arousals that release stress hormones such as catecholamine and cortisol in the blood stream. Further it induces annoyance, affects health-related quality of life, reduces sleep quality, and has been shown to affect physical activity levels. All these effects are expected to increase the incidence and progression of hypertension and myocardial infarction (MI).

Paragraph 3:

Intra-urban air pollution, particularly traffic-related air pollution, has been associated with adverse health effects in children and adults, such as cardiovascular and respiratory disorders as well as overall mortality. The World Health Organization (WHO) estimates that air pollution is responsible for approximately 7 million deaths worldwide every year. In 2012, ambient air pollution from particulate matter contributed to about 3 million deaths and 85 million disability adjusted life years globally, of which 600,000 deaths occurred yearly on the African continent. Accurate and regular air quality monitoring is necessary to evaluate air quality to determine exceedances, identify potential sources, improve control, and advise policy makers. In South Africa, air quality is monitored on a regular basis in several cities that conform to the Air Quality Management (AQM) and introduced in the Western Cape by the Department of Environmental Affairs as a measure for air quality control and planning. The first phase of this plan reported generally good air quality. However, high spatial heterogeneity was reported with poor air quality at times, especially in relation to industrial areas, high traffic conditions, and low-income residential areas.

Paragraph 4:

One quarter of the world's population is estimated to be infected with *Mycobacterium tuberculosis* (Mtb). Transmission of Mtb occurs when uninfected persons inhale infectious droplet nuclei from the infected. Droplet nuclei with diameters of $1\text{e}5$ mm can remain suspended in air for many minutes to hours, thus making Mtb highly transmissible in overcrowded locations with poor ventilation. For this and other reasons, tuberculosis (TB) remains a major public health problem worldwide. One way to estimate the risk of TB transmission begins with measuring

environmental levels of carbon dioxide (CO₂) levels. CO₂ levels combined with social contact data allow calculation of the volumes of rebreathed air in order to estimate the potential for airborne disease transmission.

Paragraph 5:

Zinc deficiency may be widespread among children in developing countries, along with high rates of serious infectious diseases such as diarrhoea, pneumonia, and malaria. Because zinc deficiency has been associated with diarrhoea and impaired immunity, recent studies have evaluated possible therapeutic or preventive roles for supplemental zinc. These studies were conducted in a variety of settings in children who were not known to be zinc deficient, but who had degrees of undernutrition typically found in developing countries. A review of these studies provides preliminary evidence that zinc deficiency is related to the incidence or outcome of serious childhood infectious diseases, and that zinc can have therapeutic or preventive benefits.

Paragraph 6:

Routine zinc supplementation in children in developing countries also appears to reduce the incidence of acute diarrhoea and possibly also the incidence of persistent diarrhoea and dysentery, at least in some subgroups. Zinc supplementation may also reduce the incidence of acute lower respiratory infections, but with one exception, studies have been of insufficient size or inappropriate design to evaluate the effects on these infrequent severe illnesses. The study from India suggesting a large effect on acute lower respiratory infections needs to be confirmed in other settings. Two studies, especially the recent one in Papua New Guinea, suggest that zinc supplementation reduces the incidence of clinical attacks of malaria in children. This finding also needs to be confirmed. Possible therapeutic effects of zinc in pneumonia and malaria also need to be evaluated. The magnitude of the therapeutic and the preventive effects of zinc on serious childhood infectious diseases appears to compare favourably with other health interventions being implemented in developing countries to improve child health and survival

Instructions for authors:

This is a reviewed assignment: follow the review process and naming conventions as explained. **The use of Artificial Intelligence tools are NOT permitted for this assignment.**

1. **Paraphrase and summarize 3 paragraphs** from epidemiological journal articles given in the script. The summary paraphrases should be about one third to half the length of the original paragraph, but contain the essential information found in the original text, stated **in your own words**.
2. Read **3** articles that will form part of your literature review. Record sources and make notes about the content. Make a literature synthesis table to record the sources. (See the example on the slides or below) OR Write a source list with 3 notes about each source. **Examples are given** in the document 'Instructions for assignments'. This part of the assignment is **not** reviewed.
3. Review the paraphrase section of the assignment written by your review partner. Follow the review process, keeping to the due dates and following the assignment naming conventions.

This assignment does not have to go through Turnitin.

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6. Bibliography

Bailey, S. 2015. *The essentials of academic writing for international students*. New York: Routledge.

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